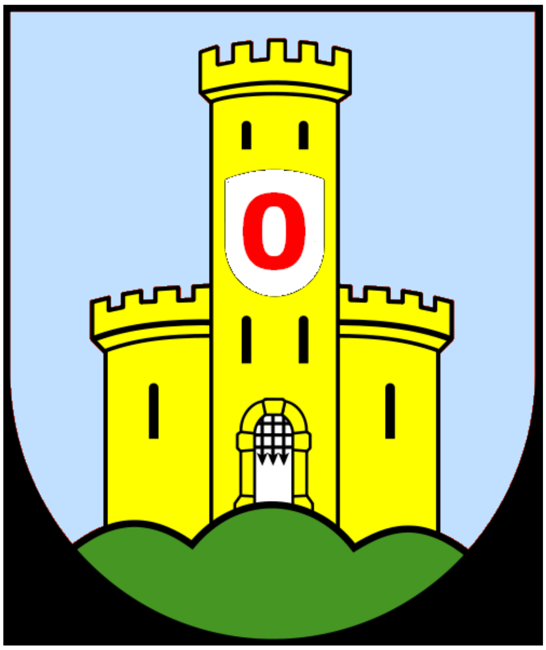
Task 2. LinearWorld Royal Tour

Available marks: 17

LinearWorld has only one dimension, so its towns are strung out in a straight line. However people can travel between any two towns by warp drive, which is thought to use a theoretical “second dimension” that physicists whisper about in corridors.

The King of LinearWorld makes an annual tour of his subjects. The chancellor has organised an itinerary for the king to visit each town exactly once. She has provided a map, comprising a list of positive integers. The integers represent the warp distance from each town to the next in sequence. Unfortunately the map doesn’t indicate which direction to travel, and the chancellor has absconded with the crown jewels. Nobody else in the kingdom is smart enough to work out the directions.



Your Task

Write a program that displays the King’s itinerary in town order. Towns are numbered from left to right, starting with 0, which is the castle and the start and end of the tour. Each other town is visited once. At each town you must decide whether to move left or right by the warp distance (number of towns) indicated on the map.

Example

Map (input)	1 4 3 2 1 3
Itinerary (output)	0 1 5 2 4 3 0

Format

- There can be up to 100 towns.
- Each data file contains several maps, each representing a new problem. The supplied data files are **task2short.dat** (up to 32 towns, 5 test cases) and **task2long.dat** (4 test cases).
- Not that it matters, but the itineraries include some well-known permutations in computer science, such as Quicksort’s worst-case input order using the median-of-three partition, pre- and post-order traversals of a complete balanced binary search tree, node order in a heap data structure, quadratic probe sequence in a hash table, and the Fibonacci sequence using modular arithmetic.
- For each problem, determine how many valid itineraries there are and display that number. If there is at least one valid itinerary, display the “first” of them, defined as the one that visits the town with the smallest available number first whenever there is a choice.
- There is a time limit of 30 seconds to complete either file.

Marking Scheme

2 marks are deducted from the available 17 for each map in either file not processed within the 30 second limit, or for an incorrect itinerary. A further **3 marks** may be deducted if any of the the itinerary counts are incorrect.

Reference: derived from part of a Stanford University programming assignment:
Zelenski, J (2008). *Assignment #3: Recursion*. see.stanford.edu/materials/icspacs106b/H18-Assign3RecPS.pdf Last accessed: 2013-08-19.